CUSTOMER SHOPPING BEHAVIOR ANALYSIS

End-to-End Data Analytics Portfolio Project

Python Data preparation & exploratory analysis	**PostgreSQL / SQL** Business questions & customer analysis
Power BI Interactive dashboard reporting	**Business Insights** Recommendations for retail decisions

PROJECT SNAPSHOT

Dataset size	3,900 purchase records	**Columns**	18
Missing data	37 values in Review Rating	**Core tools**	Python, SQL, Power BI

Prepared as a professional project report based on the supplied project document. Numerical business outcomes are not invented where the source report does not provide the actual query results.

1. Executive Summary

This project explores retail customer shopping behavior using transactional data from 3,900 purchases. The workflow combines Python-based data preparation and exploratory analysis, PostgreSQL queries for business questions, and a Power BI dashboard to communicate patterns to business stakeholders.

The analysis is designed to investigate customer spending, purchase preferences, subscription behavior, discount use, shipping choices, product ratings, and repeat-purchase patterns. The intended outcome is a clearer evidence base for customer retention, product promotion, discount strategy, and targeted marketing.

Project scope at a glance

Area	Work completed / described in source report
Data preparation	Loaded data with pandas; reviewed structure and summary statistics; handled missing review ratings; standardized column names; engineered age groups and purchase-frequency features.
SQL analysis	Used PostgreSQL to examine revenue, discount behavior, product ratings, shipping type, subscription status, customer segments, repeat buyers, and age groups.
Visualization	Created an interactive Power BI dashboard to present analysis visually.
Recommendations	Consider subscription benefits, loyalty rewards, discount-policy review, product promotion, and targeted marketing.

2. Dataset Overview

The supplied report describes a dataset with 3,900 rows and 18 columns. Its fields cover customer demographics, purchase details, and shopping behavior.

Feature group	Examples of fields	Analysis purpose
Customer profile	Age, Gender, Location, Subscription Status	Compare segments and understand customer mix
Purchase details	Item Purchased, Category, Purchase Amount, Season, Size, Color	Explore product preferences and spending
Shopping behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type	Study promotions, loyalty, ratings, and delivery choices

Data-quality note: The source report identifies 37 missing values in *Review Rating*. These were imputed using the median rating within each product category. The report also describes checking whether *Discount Applied* and *Promo Code Used* were redundant, then dropping *Promo Code Used*.

3. Methodology & Workflow

Step 1 — Load and inspect data

Use pandas to load the dataset, inspect data types and non-null counts with `df.info()`, and review numeric summaries with `df.describe()`. These checks help identify missing values, unexpected data types, and basic distribution patterns.

Step 2 — Clean and prepare

Check missing values, impute missing Review Rating values using the category-level median, standardize column names to snake_case, and review potential duplicate or redundant fields. Keep a clear record of each transformation so the analysis can be reproduced.

Step 3 — Engineer useful features

Create an `age_group` feature by grouping customer ages and a `purchase_frequency_days` feature from the purchase-frequency data, as described in the source report. Confirm the exact bin edges and frequency mapping in the notebook before publishing the final repository.

Step 4 — Load into PostgreSQL

Connect the cleaned pandas DataFrame to PostgreSQL and load it into a database table for repeatable SQL analysis. Document the table name, data types, and the process for setting up the database locally.

Step 5 — Query business questions

Use SQL aggregations, filters, grouping, ranking, and customer segmentation to answer the business questions listed in the next section. Validate each query against the intended business definition.

Step 6 — Build and explain the dashboard

Use Power BI to present key metrics and comparisons. Add clear titles, meaningful labels, consistent number formats, and filters that allow a reviewer to explore the dataset. The supplied report confirms a dashboard was built but does not include enough readable detail to reproduce its exact layout.

4. SQL Business Questions

1. Revenue by gender — compare total revenue for male and female customers.

2. High-spending discount users — identify discount users whose purchase amount exceeds the overall average.

3. Top five products by rating — rank products by average review rating.

4. Shipping comparison — compare average purchase amount for Standard and Express shipping.

5. Subscribers vs. non-subscribers — compare average spend and total revenue by subscription status.

6. Discount-dependent products — find five products with the highest percentage of discounted purchases.

7. Customer segmentation — classify customers as New, Returning, or Loyal based on purchase history.

8. Top three products per category — identify leading products within each category.

9. Repeat buyers and subscriptions — test whether customers with more than five purchases are more likely to subscribe.

10. Revenue by age group — calculate each age group's total revenue contribution.

> **Important reporting practice: the supplied source lists these questions but does not provide all final query outputs or exact numerical findings. Add the actual result tables or figures from your SQL notebook before claiming that one group outperformed another.**

5. Business Recommendations

The following actions are aligned with the recommendations in the supplied report. They should be prioritized and validated using the actual query results and dashboard metrics.

Grow subscriptions

Promote subscriber benefits and evaluate whether subscription status is associated with higher spend, revenue, or repeat purchases.

Build customer loyalty

Use a loyalty program to reward repeat customers and encourage returning customers to become loyal customers.

Review discount policy

Assess whether discounts increase purchasing activity enough to justify their impact on profit margins. Revenue alone is not the same as profit.

Promote strong products

Feature top-rated and best-selling products in campaigns, while checking both review volume and sales volume to avoid overvaluing a small sample.

Target marketing carefully

Use age-group revenue and shipping preferences to inform campaigns, then validate campaign results with measurable KPIs.

6. What This Project Demonstrates

• Data cleaning and missing-value treatment with pandas.

• Feature engineering and exploratory data analysis.

• Relational database integration and SQL business analysis.

• Analytical thinking: translating business questions into measurable queries.

• Dashboard communication with Power BI.

• Turning analysis into practical recommendations.

7. Suggested Improvements Before Publishing

• Include a concise README with project objective, dataset source, setup instructions, and screenshots.

• Add actual SQL result tables and at least three quantified findings; do not invent values.

• Include dashboard screenshots and explain the key visuals and filters.

• Add a data dictionary and document the meaning of engineered features.

• Record reproducible steps for Python dependencies and PostgreSQL setup; remove passwords, tokens, and private connection details.

• If possible, add a business KPI such as revenue, average order value, repeat-purchase rate, or subscription rate, clearly defining its formula.

8. Limitations & Interpretation

The dataset is observational and contains 3,900 purchase records, so patterns should be described as associations rather than proof of cause and effect. For example, a relationship between subscription status and spending does

not by itself prove that subscriptions cause higher spending. The supplied report does not show the final numerical results for each SQL question, so this version intentionally avoids unsupported numeric conclusions.

9. Conclusion

This project presents a practical analytics workflow from data preparation to SQL analysis and dashboard reporting. With reproducible code, transparent business definitions, actual result figures, and a well-documented GitHub repository, it can serve as a useful portfolio example for data analyst roles.

Portfolio positioning	Customer behavior analytics | Python | PostgreSQL / SQL | Power BI | Business recommendations

Source note: Prepared from the user-provided “Customer Shopping Behavior Analysis” PDF. This document reorganizes and professionally presents the supplied project scope; it does not independently verify the dataset or query outputs.